

Role: You are an expert Python developer specializing in industrial automation and GUI design using customtkinter and the pycomm3 library.

Objective: Create a professional-grade Python GUI to control a Masterflex REGLO Pump (Model 78018-xx) via EtherNet/IP.

Connection Details:

- **Protocol:** EtherNet/IP.
- **IP Address:** [Insert your Pump's IP here].
- **Interface:** Ethernet cable to PC.

Core Functionality Requirements:

1. **Connection Management:** A sidebar to input the IP address and a "Connect/Disconnect" status indicator.
2. **Operational Controls:**
 - **Start/Stop/Pause:** Toggle buttons that write to **Byte 0, Bits 0 & 1**.
 - **Direction:** A switch to toggle between CW and CCW (**Byte 0, Bit 6**).
 - **Mode Selection:** Radio buttons to switch between Continuous, Time, and Volume modes (**Byte 4**).
3. **Parameter Setting (Input):**
 - **Flow Rate:** A slider and numerical entry box (Float) writing to **Bytes 8-11**.
 - **Dispense Volume:** Input field for Volume Mode (Float) writing to **Bytes 12-15**.
 - **Batch Total:** Input field for the number of cycles (**Bytes 24-27**).
4. **Real-Time Monitoring (Feedback):**
 - **Cumulative Volume:** Display a running total from **Bytes 8-11 (Input Data)**.
 - **Current Flow Rate:** Display live feedback from **Bytes 36-39 (Input Data)**.
 - **Pump Status:** A "Traffic Light" indicator using bits from **Bytes 0-3** (e.g., Bit 1: Running, Bit 4: Head Open).
5. **Safety & Errors:**

- Display a text console that decodes Error Codes from the pump (e.g., Error #6: Head Open, Error #7: Over Temp).

UI Style: Use a dark-themed, modern dashboard aesthetic with "Masterflex Orange" accents.

Why this works

- **Direct Register Access:** By telling the AI to use pycomm3 and providing the specific byte offsets from your manual (e.g., **Bytes 8-11** for flow rate), you ensure the Python script talks directly to the pump's brain.
- **Feedback Loops:** The prompt includes "Input Data" (pump to PC) and "Output Data" (PC to pump), allowing the GUI to show you what the pump is actually doing in real-time.
- **Error Handling:** Including the error definitions from Section 5 ensures your GUI doesn't just "crash" if something goes wrong; it will tell you "Pump Head Opened" or "Motor Over Speed".

Vital Setup Steps Before Running the Code

1. **Enable the Protocol:** On the pump touchscreen, navigate to **Settings** and ensure **EtherNet/IP** is toggled to **ON**.
2. **Static IP:** In **Ethernet Settings**, switch from DHCP to **STATIC** and assign an IP (e.g., 192.168.0.50) so the Python script can find it every time.
3. **Install Libraries:** You will need to run `pip install pycomm3 customtkinter` on your computer.